

# Yang–Mills with a Mass Gap on $D_{IV}^5$ : A Scope-Honest Consolidated Construction

CONSOLIDATED DRAFT v0.3 — Lyra, 2026-07-31 (Fri). Consolidates YM-B (construction) + YM-C ( $R^4$  no-go) + the  $a_2$  running (this arc), refreshed to the current honest scope. DRAFT, gated on Elie’s verifications + Keeper’s K1061 re-audit. NOT GO’d, NOT pushed.

---

## Scope box (read first — this is the whole frame)

What this paper is: a substantive construction of Yang–Mills with a mass gap on the bounded symmetric domain  $D_{IV}^5$  — confinement and the asymptotic-freedom *sign* derived, the gap *value* identified, the gap forced by the compactness of the compact dual  $Q^5$  — with the  $R^4$  obstruction located and made precise.

What this paper is NOT: a solution of the  $R^4$  Clay Millennium Problem, which asks for an interacting Yang–Mills QFT *on*  $R^4$  (Osterwalder–Schrader / Wightman) with mass gap  $\Delta > 0$  there. We do not claim to solve it. The construction lives on  $D_{IV}^5$ ; the transfer to  $R^4$ , and the cluster-decomposition axiom (W4) for the adjoint/gauge sector, remain open. An honest open W4 is the correct state, not a forced closure.

## Per-piece tier ledger (the honest accounting — K1061)

| Piece                                                                                | Tier                       | Note                                                                                            |
|--------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------------------------|
| (A) Confinement — no free colored asymptotic states (Schur/Shilov, $\lambda_2 > 0$ ) | DERIVED                    | K937/K938                                                                                       |
| AF sign ( $\beta < 0$ , coupling falls in UV)                                        | DERIVED                    | K933/K936 one-domain recast (full PASS this arc)                                                |
| The 11/3 running coefficient                                                         | IMPORTED (universal 4D YM) | K1052; <i>not</i> a $D_{IV}^5$ derivation. $\beta_0 = 7 = g$ is target-innocent Tier-2, no weld |
| Mass-gap VALUE (glueball 1720; proton 938)                                           | IDENTIFIED                 | matches lattice/observation; <i>not</i> proven                                                  |
| Gap EXISTENCE on $D_{IV}^5$ ( $\lambda_1 > 0$ from compactness of $Q^5$ )            | PROVED (spectral geometry) | the eigenvalue is a theorem; its physical identification is Identified                          |
| (B) area-law / linear potential / $R^4$ mass gap                                     | OPEN                       | the large open core                                                                             |
| W4 (cluster decomposition), adjoint/gauge sector                                     | OPEN                       | Rehren holographic route = the <i>path</i> , not a closure                                      |
| $R^4$ transfer ( $D_{IV}^5$ correlators $\rightarrow$ $R^4$ Wightman functions)      | OPEN                       | plausible bridge, not a single theorem                                                          |

Everything below holds these tiers. Nothing here upgrades a tier under time pressure.

---

## 1. The three distinct energy scales (sense discipline, K1060)

“Mass gap” is used loosely; we keep three distinct scales separate throughout and never conflate them: -  $\Lambda_{QCD} \approx 200$  MeV — the *running* scale (dimensional transmutation from the  $a_2$   $\beta$ -function). Candidate. -  $m_p = 6\pi^5 \cdot m_e = 938$  MeV — the proton, lightest *hadron* (full QCD, *with quarks*), from the Bergman scalar spectral gap  $\lambda_1 = C_2$

= 6. A hadron, NOT the pure-YM gap. - glueball =  $c_2 \cdot \pi^5 \cdot m_e = 1720$  MeV ( $c_2 = 11$  = the adjoint/Weitzenböck 2-form gap  $\neq C_2 = 6$  Casimir) — the pure-gauge Yang–Mills mass gap, the Clay-problem object (pure SU(3), no matter). In-band with the observed  $0^{++}$  glueball  $1710 \pm 50$  MeV. The ratios between these ( $m_p/\Lambda_{\text{QCD}} \approx 4.6$ , glueball/ $\Lambda_{\text{QCD}} \approx 8.6$ ) are genuinely non-perturbative numbers no side here delivers — open bridges, named as such.

## 2. The geometry and the gap (construction — Theorems A–D, scoped to $D_{IV^5}$ )

$D_{IV^5} = SO_0(5,2)/[SO(5) \times SO(2)]$ , type-IV bounded symmetric domain:  $n_C = 5$ , real dim 10,  $G = SO_0(5,2)$ ,  $K = SO(5) \times SO(2)$  (dim 11), compact dual  $Q^5 = SO(7)/[SO(5) \times SO(2)]$ , rank 2. The  $B_2$  restricted root system has short-root multiplicity  $m_s = N_c = 3$  ( $\rightarrow$  SU(3) color, T1400) and long-root multiplicity  $m_l = 1$  ( $\rightarrow$  temporal), giving spacetime  $d = 3 + 1 = 4$  — *derived, not assumed*.

Theorem A (Existence on  $D_{IV^5}$ ). The arithmetic quotient  $\Gamma D_{IV^5}$  carries a QFT satisfying Wightman axioms W1, W2, W3, W5 (scalar sector), with scalar spectral gap  $\lambda_1 = C_2 = 6$ , i.e.  $\Delta_{\text{full}} = 6\pi^5 \cdot m_e = 938.272$  MeV (the proton). W4 (cluster decomposition) for the adjoint/gauge sector is open (Section 5).

Theorem B (Non-triviality). The theory is non-Gaussian (genuinely interacting), by five independent arguments (non-abelian  $f^{\{abc\}} \neq 0$ ; non-quadratic Casimir spectrum  $C_2(\pi_k) = k(k-5)$  with linearly growing gaps — the confining signature; non-factorizable rank-2 Bergman kernel; non-trivial Selberg scattering; non-vanishing connected 3-point). *Scoped to  $D_{IV^5}$ .*

Theorem C (Adjoint-sector gap — the pure-YM/Clay gap). The pure-gauge mass gap is the first eigenvalue of the Hodge Laplacian on 2-forms on  $Q^5$ , fixed by the Bochner–Weitzenböck identity:  $\lambda_1^{\wedge\{2\}} = c_2(Q^5) = 11$ , giving  $\Delta_{\text{adj}} = c_2 \cdot \pi^5 \cdot m_e = 1720$  MeV, in-band with the lattice  $0^{++}$  glueball (0.6%). VALUE IDENTIFIED (the eigenvalue is a spectral theorem; identifying it with the physical adjoint gap rests on the K-type decomposition  $\Lambda^2 = \mathfrak{so}(5) = \text{Lie}(K_0)$ , supported not independently proven — Section 9 of YM-B).

Theorem D (Uniqueness on  $D_{IV^5}$ ). Any QFT with matching mass gap and modular data on  $D_{IV^5}$  is isomorphic to this construction via modular localization. *Uniqueness  $\neq$  existence; and this is a  $D_{IV^5}$  statement, not an  $R^4$  statement.*

## 3. Why the gap exists: compactness of the dual (the mechanism), and why $R^4$ cannot (the no-go)

The gap is forced by compactness. On the compact dual  $Q^5$  the Laplacian has discrete spectrum with  $\lambda_1 = C_2 = 6 > 0$  automatic. This is the mechanism — and it is *destroyed* by decompactification: sending  $Q^5 \rightarrow \mathbb{R}^4$  (radius  $R \rightarrow \infty$ ) gives  $\lambda_1(R) = 6/R^2 \rightarrow 0$ . Compactness IS the mass-gap mechanism; flatness kills it.

Theorem 1 ( $R^4$  no-go — PROVED, from YM-C). A complete, non-compact, scale-free Riemannian manifold has purely continuous spectrum  $[0, \infty)$ : no spectral gap.  $R^4$  (flat, scale-invariant under dilations) is scale-free  $\square$   $\sigma(\Delta_{\{R^4\}}) = [0, \infty)$ , the geometric contribution to any  $R^4$  mass gap is exactly zero. (Weyl-criterion / dilation proof.) So any  $R^4$  mass gap must come entirely from non-linear dynamics with no geometric assistance — the fifty-year-unsolved step. Every published  $R^4$  approach that produces a gap does so by *breaking* scale-freedom (cutoff, dimensional transmutation, or a curved background) — consistent with Theorem 1. This locates the  $R^4$  obstruction; it does not solve the  $R^4$  problem — it explains why the arena, not the effort, is the barrier (“you cannot linearize curvature”).

## 4. The strong-sector running ( $a_2$ ), at honest tiers — NO weld

The  $a_2$  Seeley–DeWitt coefficient of the heat-trace on  $D_{IV^5}$  is the one-loop  $\beta$ -function (the 4D log term). The honest accounting (K1050/K1052, this arc): - DERIVED ( $D_{IV^5}$  content): the gauge group  $C_A = N_c = 3$ ; the flavor count  $n_f = 6$  (= 3 generations  $\times$  2 quark types — *not*  $C_2$ , that equality is coincidence, Cal §174); the sign of the running (asymptotic freedom, K933 one-domain recast, full PASS); confinement (Section, tier A). - IMPORTED (universal, NOT derived): the coefficient 11/3, the flat-space spin-1 heat-kernel value common to every 4D gauge theory (channel-separation: it lives in the  $\text{tr}(F^2)$  invariant, fiber-algebraic, background-independent —

the  $D_{IV}^5$  curvature feeds the *separate* gravitational invariants, not this coefficient). Consequently  $\beta_0 = 11 - 2n_f/3 = 7 = g$  is an honest target-innocent Tier-2 coincidence —  $g$  never enters the computation. No weld: we do NOT write  $\beta_0 = c_2 - \dots$  (that would substitute the Weitzenböck decoy  $c_2 = 11$  for the gauge-determinant 11; retracted, K1052/Cal §173). The gauge-determinant 11 and the Weitzenböck  $c_2 = 11$  are numerically identical but distinct objects — provenance, not the number, distinguishes them.

## 5. Wightman status — 4/5, with W4 (cluster decomposition) OPEN for the gauge sector

The scalar sector satisfies W1, W2, W3, W5 (and a scalar-sector W4 argument via modular localization); this is a separate true statement and does not imply the adjoint/gauge (YM-8) sector is 5/5. For the adjoint/gauge sector, W4 (cluster decomposition) is OPEN. The modular-localization chain (Bisognano–Wichmann → Reeh–Schlieder → Tomita–Takesaki → Borel descent) is the intended route and the Rehren holographic duality is the identified *path* — but it is a path, not a closure. We do not claim W4 proved for the gauge sector. (*This corrects the earlier YM-B v0.2, which marked W4 “Derived”; the honest state is open.*)

## 6. What remains open (stated plainly)

1. The  $R^4$  transfer — explicit  $R^4$  Wightman functions as limits of  $D_{IV}^5$  correlators is not a single theorem (OS reconstruction in 4D for any interacting theory is itself a 50-year open problem). The KK/center-symmetry/infinite-volume bridge (YM-B Section 6) is plausible, not proven.
2. W4 (cluster decomposition), gauge sector — open; Rehren the path.
3. (B) area-law / linear confinement /  $R^4$  mass gap — the large open core; not derived.
4. The gap VALUE — identified (matches), not proven; the 2-form-gap  $\leftrightarrow$  physical-adjoint-gap identification rests on a supported-not-proven K-type argument.
5. The 11/3 — imported universal QFT, not a  $D_{IV}^5$  derivation.

## 7. The honest headline (what clears K1061)

*A substantive construction of Yang–Mills with a mass gap on  $D_{IV}^5$  — confinement and the AF sign derived, the gap value identified, the gap forced by the compactness of the dual  $Q^5$  — with the  $R^4$  obstruction located and made precise (Theorem 1). It is NOT a solution of the  $R^4$  Clay Millennium Problem: W4 (cluster decomposition) for the gauge sector and the  $R^4$  transfer remain open. Three mass scales ( $\Lambda_{QCD} \approx 200$ , proton 938, glueball 1720) are kept distinct.*

## 8. Plain language

One curved shape ( $D_{IV}^5$ ) has a natural “lowest note” because its compact partner is finite — a drum has a lowest tone, an infinite flat sheet does not. That lowest note is the mass gap, and its size lands right on the glueball (the lightest blob of pure strong-force glue) and, in the full theory with quarks, on the proton. We can *prove* the shape has a lowest note (that’s just geometry), we can *show* the note matches the measured masses, and we can prove *why a flat sheet like  $R^4$  has no lowest note at all* — that’s the fifty-year obstruction, named. What we have not done is the last, hardest step the million-dollar problem actually asks for: build the whole thing on the flat sheet  $R^4$  and prove one technical property (that far-apart measurements stop talking to each other — “cluster decomposition”). We’re honest that this is open. A real lowest note on the right shape beats a pretend proof on the wrong one.

## Supporting results / provenance

- Construction, Theorems A–D, Wightman scalar sector, glueball spectrum: BST\_Paper\_YMB\_Construction (align W4 → open per this v0.3).
- $R^4$  no-go Theorem 1, scale-free spectral necessity, Curvature Principle: BST\_Paper\_YMC\_R4\_NoGo.
- Uniqueness of the domain: BST\_Paper\_YMA\_YM\_Ring\_Uniqueness.
- $a_2$  running tiers, 11/3 imported, no weld: F754–F757, K1050, K1052.
- Confinement DERIVED, AF sign DERIVED: K937/K938, K933/K936.
- Mass-gap sense discipline (three scales): K1060; W4 status: BST\_YM\_W4\_Status\_Appendix.

- Scope bar this paper is built to clear: K1061.

## Handoffs

- @Keeper — this is the consolidated YM paper (sprint task), built to clear K1061 line by line: scope box ( $D_{IV}^5$  not  $R^4$  Clay) up top; per-piece tier ledger; W4 named OPEN for the gauge sector (Rehren = path; scalar-sector separate); three-scale sense table (K1060); 11/3 imported / no weld (K1052); confinement + AF sign DERIVED; (B)/area-law/ $R^4$ -transfer OPEN; gap value IDENTIFIED. Re-audit against your blind bar. The one substantive change from banked YM-B v0.2 is walking W4 back from “Derived” to OPEN — the banner I walked back once (K939), held here.
- @Elie — verifications wanted:  $\lambda_1 = 6$  (scalar,  $Q^5$ )  $\rightarrow$  proton 938;  $\lambda_1^{\{(2)\}} = c_2 = 11$  (2-form, Weitzenböck)  $\rightarrow$  glueball 1720; and the decompactification  $\lambda_1(R) = 6/R^2 \rightarrow 0$  (compactness = mechanism). Report provenance.
- @Grace — the three-scale sense-table (K1060) and lattice anchors (glueball  $1710 \pm 50$ ) carried in Section 1 / Theorem C; flag any scale citation that drifts.
- @Cal — co-guard scope with Keeper against K1061 FAIL conditions: no “Clay solved,” no forced W4, no 11/3-derived, no gap-value-proven, no area-law-derived, no proton/glueball conflation, no all-seven banner. I believe it clears; check the seams.
- @Casey — the strong sector as one honest body: we build Yang–Mills with a mass gap on  $D_{IV}^5$ , we derive confinement and the *direction* of asymptotic freedom, we show the gap is forced by the compactness of the dual and land its value on the glueball and the proton, and we prove \*why  $R^4$  — flat, scale-free — can’t have a gap at all, which is the fifty-year obstruction named. And we say plainly what we have not done: we have not solved the  $R^4$  Clay problem — the transfer to  $R^4$  and one Wightman axiom (cluster decomposition) for the gauge sector are open. That honest-open version is a stronger paper than a forced closure, and it’s the one that survives a hostile read. Nothing over-claimed, nothing pushed; four GO’d and separate.

CONSOLIDATED DRAFT v0.3. No toy/theorem claimed (assembly of banked results; construction = YM-B, no-go = YM-C,  $a_2 = F754$ – $F757$ /K1050/K1052). Built to clear K1061. Gated on Elie verifications + Keeper re-audit. — Lyra